

Solution Requirements

Date	05-07-2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can securely access the system through a simple login or authenticated interface before using prediction services.	High
2	Authorization levels	Farmers can access crop recommendations, while administrators can manage datasets, machine learning models, and system settings	High

3	External interfaces	The system connects with the trained Machine Learning model, crop dataset, and can be extended to integrate weather APIs in the future.	High
4	Transactions processing	The system accepts soil and weather parameters, processes them through the ML model, and instantly generates crop recommendations.	High
5	Reporting	Displays recommended crop results and can generate prediction history and summary reports for users.	Medium
6	Business rules, etc.	Crop recommendations are generated only after validating all mandatory soil and weather input parameters.	High
7	Compliance to laws or regulations	The system protects user information and follows general data privacy and secure data handling practices.	Medium
8	<i>Other</i>	Provides a simple, responsive, and user-friendly interface for farmers to obtain quick recommendations.	Medium

Step 2: Non-Functional Requirements (NFR)

1	Performance & Speed	The system should generate crop recommendations quickly	Prediction generated in less than 2 seconds
2	Scalability	The application should support multiple users simultaneously without significant performance degradation	Supports 100+ concurrent users
3	Security & Data Privacy	User input data should be processed securely and protected from unauthorized access.	Secure data handling with authenticated access
4	Reliability & Availability	The application should remain operational with minimal downtime.	99% system availability
5	Usability & Accessibility	The interface should be simple, responsive, and easy to use on desktop and mobile devices.	Responsive UI with easy navigation

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

6	Other	The system should be easy to maintain, update, and deploy on different platforms.	Modular code and cross-platform compatibilit
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